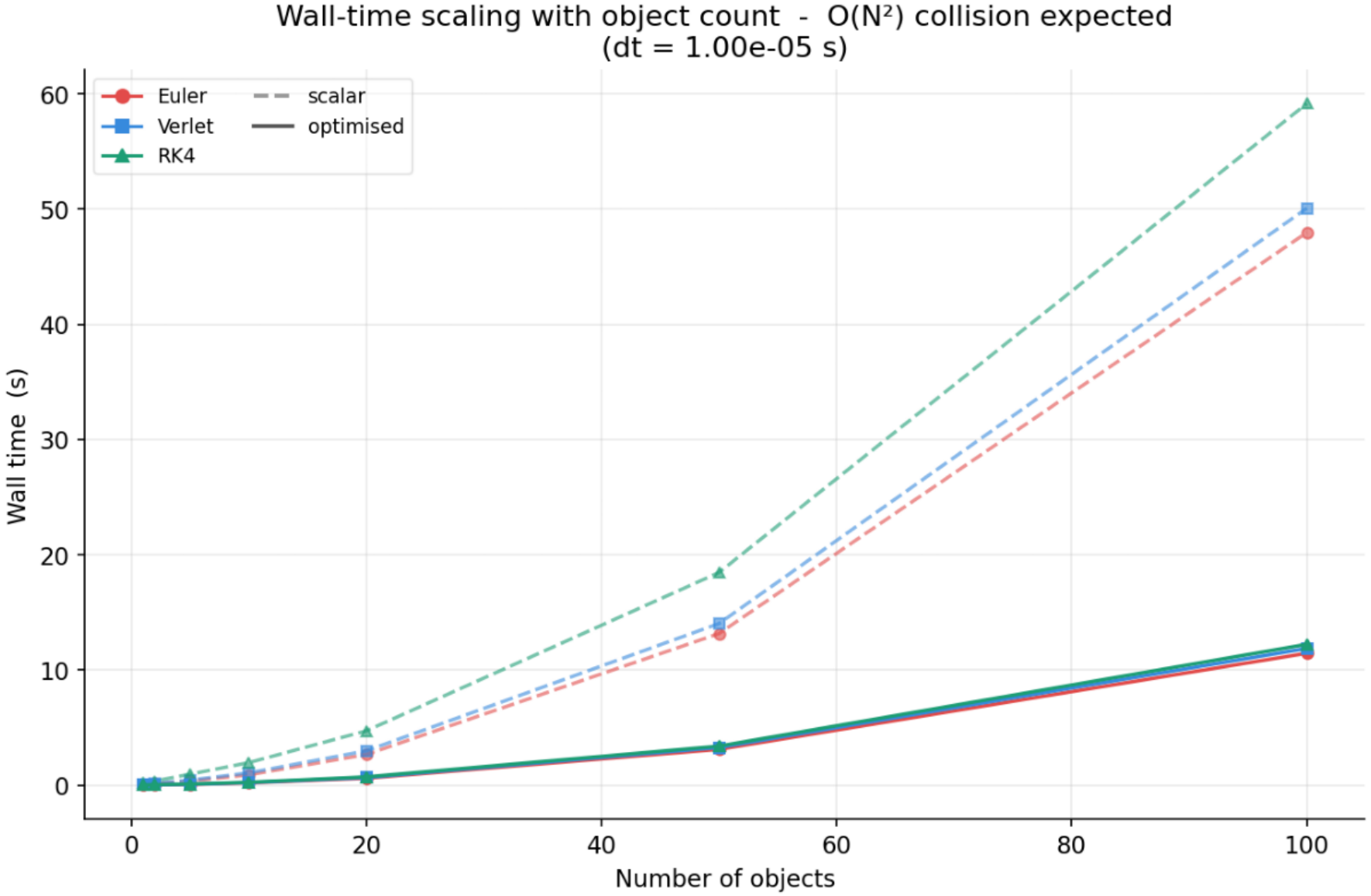
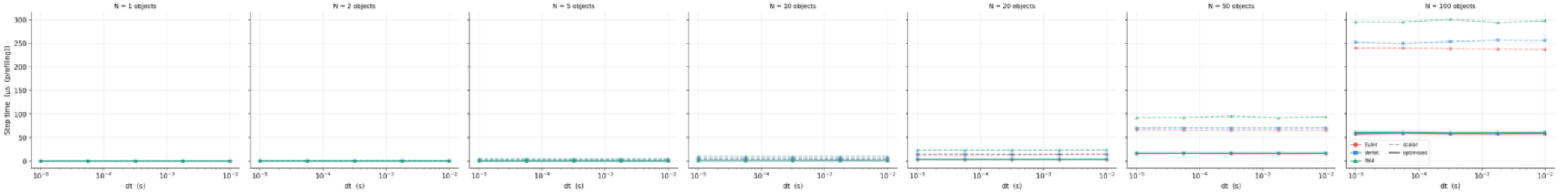
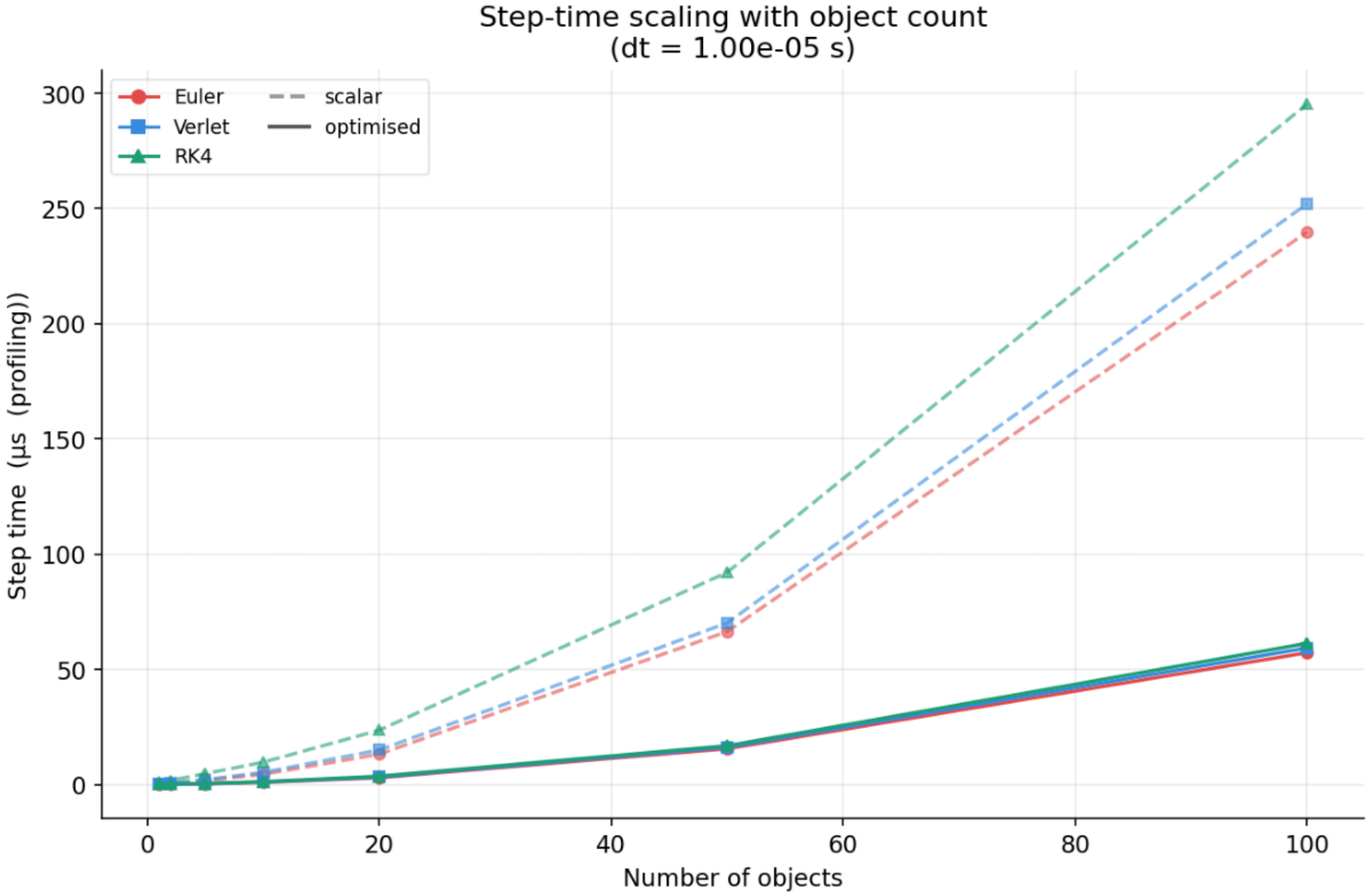
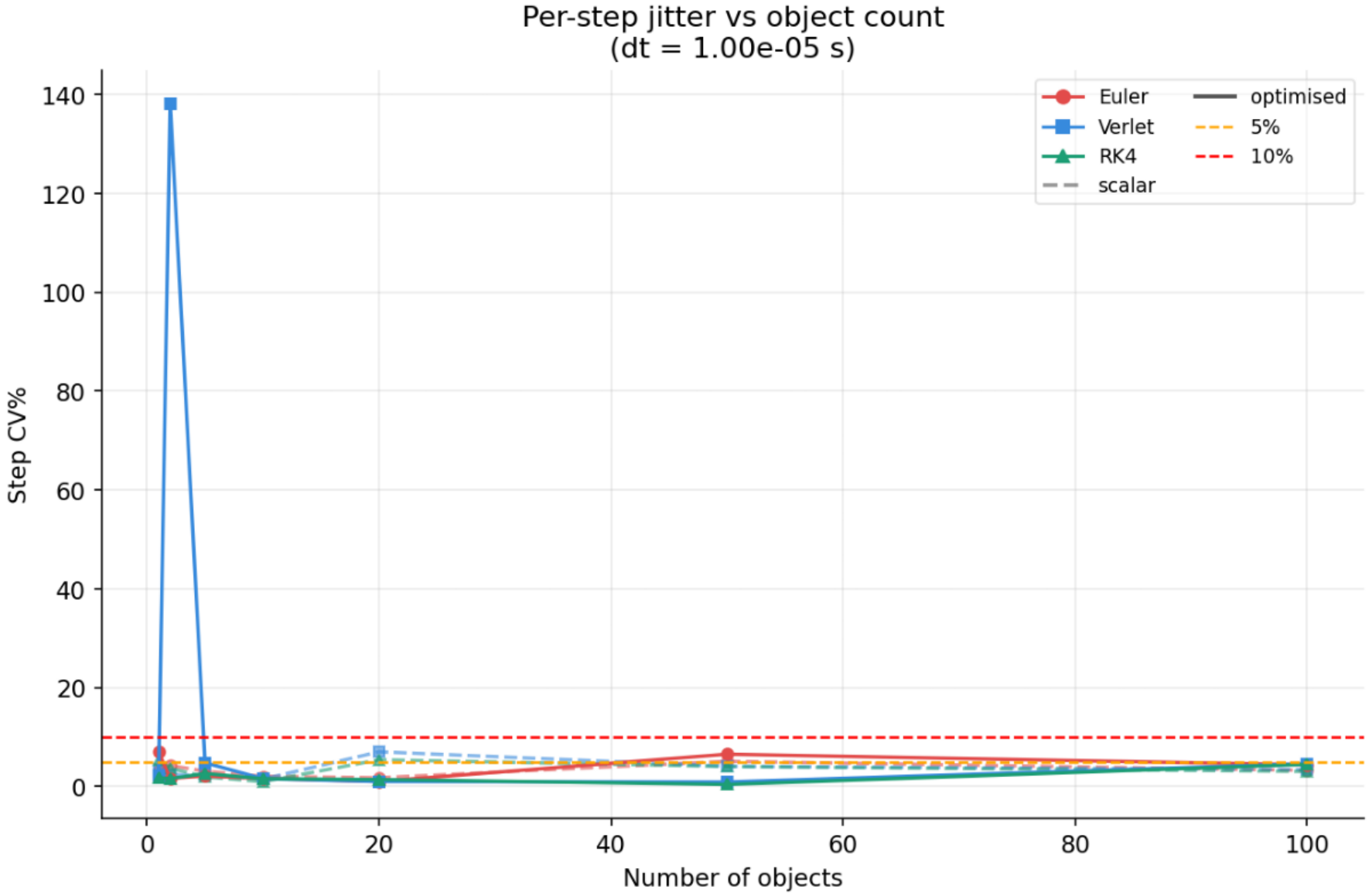

Wall time vs dt - O(1/dt) scaling expected
N = 1 objects
N = 2 objects
N = 5 objects
N = 10 objects
N = 20 objects
N = 50 objects
N = 100 objects
Wall time (s)
dt (s)
Euler
Verlet
RK4
scalar
optimised

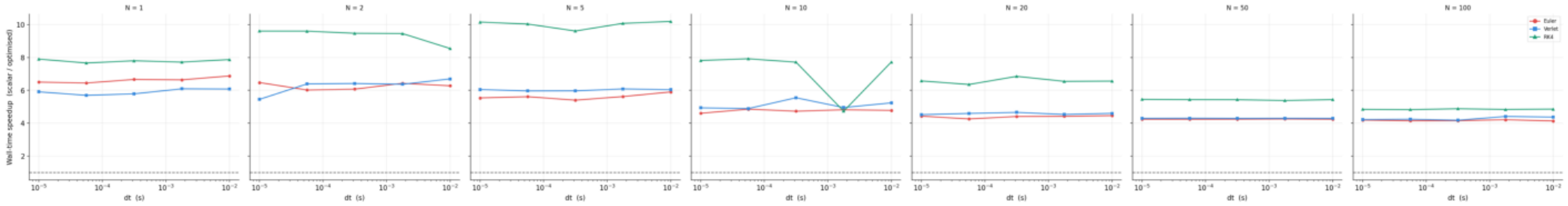

Wall-time scaling with object count - O(N²) collision expected
(dt = 1.00e-05 s)
Euler
Verlet
RK4
scalar
optimised
Wall time (s)
Number of objects
0
10
20
30
40
50
60
0
20
40
60
80
100

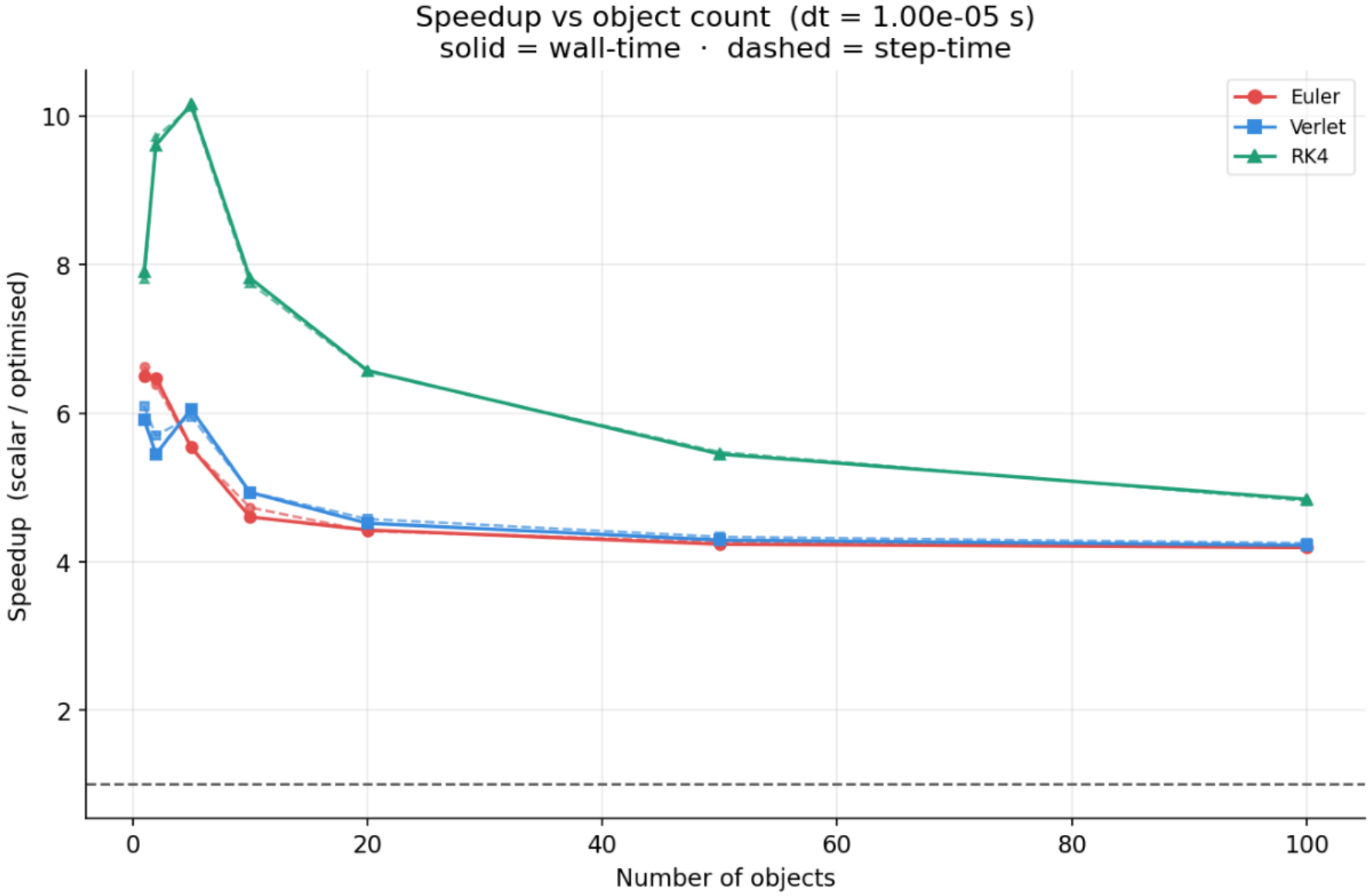

Step time vs dt
(plateau at fine dt = true per-step cost)
N = 1 objects
N = 2 objects
N = 5 objects
N = 10 objects
N = 20 objects
N = 50 objects
N = 100 objects
Step time (µs (profiling))
dt (s)
Euler
Verlet
RK4
scalar
optimised


Step-time scaling with object count
(dt = 1.00e-05 s)
Euler
Verlet
RK4
scalar
optimised
Step time (µs (profiling))
Number of objects
0
50
100
150
200
250
300
0
20
40
60
80
100

Per-step jitter vs dt (low CV → regular cost)
N = 1 objects
N = 2 objects
N = 5 objects
N = 10 objects
N = 20 objects
N = 50 objects
N = 100 objects
Step CV%
dt (s)
Euler
Verlet
RK4
scalar
optimised
5%
10% (high jitter)


Per-step jitter vs object count
(dt = 1.00e-05 s)
Euler
Verlet
RK4
scalar
optimised
5%
10%
Step CV%
Number of objects
0
20
40
60
80
100
120
140


Scalar → Optimised wall-time speedup vs dt (> 1 = optimised wins)
N = 1
N = 2
N = 5
N = 10
N = 20
N = 50
N = 100
Euler
Verlet
RK4
Wall-time speedup (scalar / optimised)
dt (s)


Speedup vs object count (dt = 1.00e-05 s)
solid = wall-time · dashed = step-time
Euler
Verlet
RK4
Speedup (scalar / optimised)
Number of objects
10
8
6
4
2
0
20
40
60
80
100